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# 1 Mathematical Background

This chapter collects the mathematical material needed later for randomized least-squares methods and for covariance-based dependence correction. The main ingredients come from four directions: classical least-squares theory, random projections and subspace embeddings, symmetric positive definite operators and their matrix functions, and perturbation questions related to conditioning. Although all spaces in this thesis are finite-dimensional, it is often convenient to use the language of linear maps, ranges, and subspaces rather than only matrix coordinates.

For an overdetermined system  $A \in \mathbb{R}^{n \times d}$  with  $n \geq d$ , the least-squares problem

$$\min_{x \in \mathbb{R}^d} \|Ax - b\|_2$$

asks for the vector whose image under  $A$  is closest to  $b$  in Euclidean norm. Geometrically, this means projecting  $b$  onto the range of  $A$ . If  $A$  has full column rank, the minimizer is unique. In exact arithmetic, it may be characterized by the normal equations or, more stably, by an orthogonal factorization.

The randomized viewpoint replaces the original problem by a smaller one. A sketching matrix  $S \in \mathbb{R}^{s \times n}$ , with  $s$  much smaller than  $n$ , is chosen so that the geometry of the range of  $A$  is approximately preserved after compression. One then solves

$$\min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2.$$

The quality of this replacement depends on how well the sketch preserves norms and angles on the relevant subspace. This is where subspace embeddings and the Johnson–Lindenstrauss phenomenon enter the picture. The sketch should act almost isometrically on the subspace that carries the least-squares geometry. The formal definitions and embedding guarantees are given in [Chapter 3](#) → p.11.

Two sketch families are especially important in this thesis. The first is the Gaussian sketch, whose entries are independent random variables with suitable scaling. The second is the class of structured sketches, especially transforms based on randomized Hadamard matrices. Gaussian sketches are the cleanest theoretical baseline. Structured sketches are more attractive computationally because they can exploit fast transforms.

Conditioning plays a central role throughout. Even when sketching gives a good approximation in principle, the numerical behavior of the reduced problem depends strongly on the conditioning of  $SA$ , that is, on the stability of the compressed operator on the relevant subspace. This becomes even more important in finite precision, since rounding errors may be harmless for a well-conditioned problem and severe for an ill-conditioned one. The relation between sketching quality and conditioning is discussed in [Proposition 3.7](#) → p.16.

The later mixed-precision discussion therefore rests on a simple principle. Randomized approximation error and floating-point error should not be studied in isolation. A useful theory must account for both at the same time.

## 1.1 Singular Value Decomposition and Pseudoinverse

The basic structural tool for least-squares analysis is the thin singular value decomposition. If  $A \in \mathbb{R}^{n \times d}$  has rank  $r$ , then

$$A = U_r \Sigma_r V_r^\top,$$

where  $U_r \in \mathbb{R}^{n \times r}$  and  $V_r \in \mathbb{R}^{d \times r}$  have orthonormal columns and

$$\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r), \quad \sigma_1 \geq \dots \geq \sigma_r > 0,$$

contains the nonzero singular values. This factorization separates the geometry of the range, the nullspace, and the spectral scales of the operator.

If  $A$  has full column rank, the least-squares minimizer may be written as

$$x_\star = A^\dagger b, \quad A^\dagger = (A^\top A)^{-1} A^\top = V_r \Sigma_r^{-1} U_r^\top,$$

where  $A^\dagger$  is the Moore–Penrose pseudoinverse. This formula is useful for analysis because it makes explicit that the small singular values of  $A$  are the source of sensitivity: directions associated with  $\sigma_j \ll 1$  are amplified by  $\sigma_j^{-1}$ .

The operator norm of  $A$  is

$$\|A\|_2 = \sigma_1,$$

and for a full-column-rank matrix the Euclidean condition number is

$$\kappa_2(A) = \frac{\sigma_1}{\sigma_r}.$$

This quantity measures how strongly data perturbations can be amplified by the least-squares solve and reappears later in the analysis of sketched systems.

## 1.2 Normal Equations, QR, and SVD

The least-squares problem admits several algebraic formulations, but they do not behave equally well numerically. The normal equations

$$A^\top A x = A^\top b$$

are attractive because they reduce the problem to a square system. Their drawback is that they square the spectrum:

$$\sigma_j(A^\top A) = \sigma_j(A)^2, \quad \kappa_2(A^\top A) = \kappa_2(A)^2.$$

Thus a moderately ill-conditioned least-squares problem can become a severely ill-conditioned linear system when written in this form.

By contrast, if  $A = QR$  is a thin QR factorization with  $Q$  having orthonormal columns and  $R \in \mathbb{R}^{d \times d}$  upper triangular, then the least-squares problem reduces to

$$\min_x \|Rx - Q^\top b\|_2,$$

or equivalently to the triangular solve

$$Rx = Q^\top b$$

in the full-column-rank case. QR avoids the explicit formation of  $A^\top A$  and is therefore the standard stable dense formulation.

The SVD is more expensive than QR but more informative. It reveals rank deficiency directly, yields the pseudoinverse explicitly, and makes clear how solution sensitivity depends on the singular spectrum. In the present thesis, QR is the practical dense baseline, while the SVD is the conceptual reference that makes the conditioning questions transparent [1].

## 1.3 Least-Squares Perturbation Viewpoint

The numerical sensitivity of least squares depends on more than the matrix norm alone. Let  $x_\star$  solve

$$\min_x \|Ax - b\|_2,$$

and write  $r_\star = b - Ax_\star$  for the optimal residual. Classical perturbation theory shows that under small perturbations  $\Delta A$  and  $\Delta b$ , the relative change in the minimizer is controlled by the condition number together with a term that reflects the size of the residual. Schematically,

$$\frac{\|\Delta x_\star\|_2}{\|x_\star\|_2} \lesssim \kappa_2(A) \left( \frac{\|\Delta A\|_2}{\|A\|_2} + \frac{\|\Delta b\|_2}{\|b\|_2} \right) + \kappa_2(A)^2 \frac{\|r_\star\|_2}{\|A\|_2 \|x_\star\|_2} \frac{\|\Delta A\|_2}{\|A\|_2},$$

with more precise constants depending on the chosen formulation [Ch. 1 [1]]. The important qualitative message is that solution sensitivity is mild when the residual is small and  $A$  is well conditioned, but can be much worse in inconsistent or ill-conditioned problems.

This observation explains why residual bounds and solution-error bounds should be kept conceptually distinct. A method may preserve the least-squares objective accurately while still producing a noticeably perturbed minimizer if the problem is geometrically sensitive. That distinction becomes central later when sketching and floating-point perturbations are analyzed together.

## 1.4 Floating-Point Recap for Least Squares

The mixed-precision chapter develops the full floating-point model later, but the least-squares background already needs one simple principle. Forming the reduced operator, the normal matrix, or the orthogonal factorization in finite precision does not solve the exact problem posed by  $A$  and  $b$ ; it solves a nearby problem. The size of this nearby perturbation depends on the working precision, the problem dimensions, and the stability of the chosen factorization. That is why the choice between normal equations, QR, SVD, and later randomized surrogates is not only a question of flop counts but also of how perturbations are amplified through the conditioning of the underlying operator [2].

## 1.5 Symmetric Positive Definite Operators

The covariance-transport part of the thesis is built from symmetric positive semidefinite and positive definite matrices. A matrix  $A \in \mathbb{R}^{p \times p}$  is symmetric positive semidefinite if

$$x^\top A x \geq 0 \quad \forall x \in \mathbb{R}^p,$$

and symmetric positive definite if the inequality is strict for every nonzero  $x$ . In the latter case,  $A$  is invertible and all of its eigenvalues are strictly positive. Since the matrices are symmetric, they admit an orthogonal spectral decomposition

$$A = Q \Lambda Q^\top,$$

where  $Q$  is orthogonal and  $\Lambda$  is diagonal with real entries.

Empirical covariance matrices provide the main examples later on. If  $Z \in \mathbb{R}^{n \times p}$  is a centered data matrix, then

$$C = \frac{1}{n} Z^\top Z$$

is symmetric positive semidefinite. In high-dimensional settings it may be singular or nearly singular, which is why regularization by  $\lambda I$  is not an optional detail but a structural part of the method.

## 1.6 Matrix Square Roots and Regularization

If  $A$  is symmetric positive definite with spectral decomposition  $A = Q \Lambda Q^\top$ , its matrix square root and inverse square root are defined by functional calculus as

$$A^{1/2} = Q \Lambda^{1/2} Q^\top, \quad A^{-1/2} = Q \Lambda^{-1/2} Q^\top.$$

These matrices are again symmetric positive definite, and they are the unique positive definite solutions of

$$A^{1/2} A^{1/2} = A, \quad A^{-1/2} A A^{-1/2} = I.$$

Regularization shifts the spectrum away from zero. If  $C$  is positive semidefinite and  $\lambda > 0$ , then

$$C + \lambda I$$

is positive definite, with eigenvalues  $\mu_i(C) + \lambda$ . This shift provides a numerical safety margin for the inverse square root and controls how strongly small covariance eigenvalues are amplified.

## 1.7 Basic Perturbation Facts

The later covariance-transport analysis repeatedly uses simple spectral perturbation facts. The first is positivity under perturbation: if  $A$  is symmetric positive definite and  $E$  is symmetric, then

$$\lambda_{\min}(A + E) \geq \lambda_{\min}(A) - \|E\|_2.$$

Consequently, if  $\|E\|_2 < \lambda_{\min}(A)$ , the perturbed matrix remains positive definite. This is the basic condition that allows the square root and inverse square root of a computed approximation to remain well defined.

The second fact is qualitative rather than exact: matrix square roots and inverse square roots inherit the conditioning of the underlying spectrum. Perturbations of  $A^{1/2}$  are controlled by the distance of the spectrum from zero, while perturbations of  $A^{-1/2}$  are more sensitive because the scalar map  $t \mapsto t^{-1/2}$  has derivative  $-\frac{1}{2}t^{-3/2}$ . This is the analytic source of the regularization tradeoff in covariance transport. Increasing  $\lambda$  improves stability of the inverse square root, but it also changes the operator that is being transported.

## 1.8 Why This Background Matters

These spectral facts extend the least-squares viewpoint from compressed rectangular operators to self-adjoint covariance operators. In the least-squares chapters, the decisive question is whether sketching preserves the geometry of a range space well enough that the reduced problem remains faithful to the original one. In the covariance-transport chapter, the decisive question becomes whether a randomized and finite-precision approximation of the covariance operator remains accurate enough after it is processed by matrix square roots and inverse square roots. The mathematical language is therefore the same in spirit, but the operator calculus is richer.

# 2 Mixed-Precision Arithmetic

Mixed-precision arithmetic means that one algorithm uses more than one floating-point format. The idea is simple. One tries to perform the expensive part of the computation in a cheaper arithmetic and keeps the more accurate format for the numerically sensitive stages. In numerical linear algebra, however, this is not only a matter of implementation. A change of precision changes the perturbation model of the algorithm itself, and therefore changes the map that is actually computed [3] [2].

This chapter introduces the basic concepts needed later for randomized least-squares solvers. In particular, we discuss floating-point representation, rounding, conditioning, and the way mixed precision interacts with algorithmic structure. The main question for the later chapters is the following: if the algorithm already contains approximation through randomized sketching, how much additional perturbation from lower precision can be tolerated?

## 2.1 Floating-Point Model

Floating-point arithmetic replaces the continuum of real numbers by a finite set of representable values. A normalized floating-point number has the schematic form

$$x = \pm m\beta^e,$$

where  $\beta$  is the base,  $m$  is the significand, and  $e$  is the exponent. Since most real numbers are not exactly representable, every arithmetic operation involves rounding. Thus finite precision is not a small correction to an exact algorithm. It is the environment in which the algorithm exists.

For rounding to nearest, the standard model writes one basic operation as

$$\text{fl}(x \circ y) = (x \circ y)(1 + \delta), \quad |\delta| \leq u,$$

where  $u$  is the unit roundoff of the working format. Lower precision means a larger  $u$ , a smaller dynamic range, and therefore a smaller safety margin against cancellation, overflow, and accumulation of rounding errors. The important point is that changing precision changes the size of the perturbation allowed at every arithmetic step. This is why low precision can be attractive computationally and dangerous numerically at the same time [2] [4].

The standard rounding model is useful because it allows us to interpret the computed algorithm as a nearby problem. Once we pass from one scalar operation to inner products, matrix-vector products, or matrix-matrix products, the perturbations are no longer controlled only by  $u$ , but also by the dimensions and the algebraic structure of the computation. For least-squares problems, this is especially important. Residuals, orthogonal factorizations, normal equations, and sketching operations all inherit their own precision-dependent perturbations.

## 2.2 Why Mixed Precision Matters

Mixed precision has become important because current hardware does not treat all precisions equally. Modern CPUs and especially accelerators often provide much higher throughput for half- or single-precision kernels than for double precision, while also reducing memory traffic and storage costs. Therefore, what used to be an implementation detail has become a design question for algorithms themselves [3] [2].

The real appeal, however, is algorithmic. Many linear algebra procedures are not equally sensitive to rounding. Bulk matrix products may tolerate lower precision, whereas factorizations, orthogonalization, residual computations, or correction solves may require a more accurate format. Mixed precision is therefore best understood as a way of deciding where accuracy is needed and where it is not. One does not choose only how much arithmetic is performed, but also where the larger perturbations are allowed to enter the computation.

## 2.3 Error Sources and Conditioning

In linear algebra computations, several error sources act at the same time. The data may already be perturbed when stored in low precision. Each arithmetic operation contributes a local rounding error. These perturbations accumulate through matrix products, factorizations, and iterative processes. Finally, the whole effect is filtered through conditioning. A well-conditioned problem may absorb the perturbation harmlessly, while an ill-conditioned one may amplify it strongly. Therefore, forward error, backward error, and conditioning must be kept distinct. Mixed precision changes the perturbation introduced by the algorithm, while the problem itself determines how much this perturbation is amplified.

This is especially important for least-squares problems. One must distinguish between the conditioning of the approximation problem itself and the conditioning of the algebra actually used to solve it. The latter may involve normal equations, QR factorizations, augmented systems, or preconditioned iterative refinement. Recent work on mixed-precision least squares shows that lower precision can be effective, but only in conditioning regimes that still allow stable refinement or correction [5] [6] [7].

## 2.4 Mixed Precision in Linear Algebra Algorithms

The classical example is iterative refinement. One performs the expensive solve in lower precision, computes a more accurate residual in higher precision, and uses this residual to correct the solution. This idea is old, but modern hardware has made it newly important. Foundational work showed that one can often recover double-precision accuracy while exploiting cheaper lower-precision factorization

steps, and more recent GPU implementations show substantial speed gains when tensor-core arithmetic is used carefully [8] [9] [2].

For least-squares problems the situation is more subtle than for square linear systems, because orthogonality, residual quality, and conditioning of augmented formulations all matter. This is why mixed-precision least-squares solvers often combine low-precision bulk work with higher-precision residuals, QR-based corrections, or Krylov refinement steps [5] [6]. The main issue is not only whether the problem can be solved cheaply, but whether the approximation to the projection onto the range of the data operator remains sufficiently accurate.

Recent work extends this picture beyond standard least squares. Mixed-precision FGMRES-based iterative refinement has been developed for weighted least squares, and related augmented-system approaches have been proposed for constrained and generalized least-squares problems [10] [7]. The common lesson is that lower precision is most useful in the bulk linear algebra that constructs an approximate solve or preconditioner, while the residual and correction stages must remain sufficiently accurate to control the final error.

## 2.5 Connection to Randomized Numerical Linear Algebra

Randomized algorithms add a second approximation before floating-point effects are even considered. A sketching matrix  $S$  replaces the original least-squares problem by a reduced one involving  $SA$  and  $Sb$ . Mixed precision can then enter at several places: in storing the data, in forming the compressed operator, in solving the reduced problem, and in any refinement step that follows.

These stages need not have the same numerical requirements. If the dominant cost lies in forming  $SA$ , low-precision matrix multiplication may be attractive. If the reduced problem is ill-conditioned, however, that gain may disappear unless the subsequent solve or residual computation is more accurate. Recent work makes this point explicit. Mixed-precision analysis of sketching-based preconditioners shows that one can choose separate precisions for the sketching and factorization stages and still obtain an effective preconditioner for GMRES-based iterative refinement, provided the perturbations remain compatible with the conditioning of the least-squares problem [11]. On the implementation side, hardware-oriented work on random projection exploits the fact that the random matrix itself may tolerate reduced precision better than the data matrix or the final solve [12].

The guiding view of this thesis is therefore the following. Sketching error and floating-point error should be analyzed together as coupled perturbations of the original least-squares problem. In the sketch-and-solve setting, one asks whether the computed reduced problem still represents the original one faithfully enough. In the sketch-and-precondition setting, one asks whether lower precision preserves

enough of the conditioning improvement for the downstream solver to remain effective.

## 2.6 Balancing Sources of Inexactness

Mixed precision is most natural when rounding is not the only source of error in the algorithm. Randomized approximation, low-rank truncation, regularization, and even finite-sample modelling choices already introduce tolerances below which additional arithmetic accuracy may not be useful. In such settings, the purpose of precision selection is not to minimize rounding error at all costs. It is to choose an arithmetic regime in which rounding remains subordinate to the inexactness already accepted elsewhere in the method. This viewpoint aligns with recent work on balancing multiple sources of inexactness rather than optimizing one numerical layer in isolation [13] [2].

This idea can be summarized schematically as

$$\eta_{\text{fp}} \lesssim \eta_{\text{alg}},$$

where  $\eta_{\text{fp}}$  denotes floating-point perturbation and  $\eta_{\text{alg}}$  the algorithmic error already built into the method. In a randomized setting, the more specific target is often

$$\eta_{\text{fp}} \lesssim \eta_{\text{rand}},$$

meaning that reduced precision is acceptable only when it does not dominate the randomized approximation error. In an application-driven setting one may relax this further to

$$\eta_{\text{fp}} \lesssim \tau_{\text{stat}},$$

where  $\tau_{\text{stat}}$  represents statistical or modelling tolerance.

This balancing viewpoint is especially important for the later covariance-transport analysis. There, the approximation error from randomized low-rank compression is already present before the matrix square roots and inverse square roots are evaluated. Computing every preceding product in the highest available precision may therefore solve one part of the numerical problem much more accurately than the later regularized transport map or the downstream application actually warrants. The aim is not maximal local accuracy, but a numerically coherent distribution of inexactness across the pipeline.

## 2.7 Role in This Thesis

Within this thesis, mixed precision is not a side topic. It is one of the main ways in which both randomized least-squares methods and randomized covariance transport are evaluated. The central question is not whether lower precision is always safe, but where it is safe, where it is useful, and where it changes the numerical character of the method too much to justify the computational gain.

The later chapters return to this question in numerical experiments and, later, in the climate-data workflow. The aim is modest but important: to understand mixed precision as part of algorithm design, not as a final implementation tweak applied after the mathematics has already been fixed.

# 3 Randomized Least Squares

Randomized numerical linear algebra uses randomness as a computational resource for restructuring expensive linear algebra tasks into smaller or more informative surrogate problems. The area now covers low-rank approximation, matrix decompositions, regression, and randomized preconditioning, but overdetermined least-squares problems remain one of its clearest and most influential successes [14] [15] [16].

This chapter focuses on sketching-based least-squares methods, which provide the first model problem for the thesis. The starting point is the overdetermined problem

$$\min_{x \in \mathbb{R}^d} \|Ax - b\|_2,$$

where  $A \in \mathbb{R}^{n \times d}$  has many more rows than columns. Interpreted geometrically, one seeks the point in the range of the linear map represented by  $A$  that is closest to  $b$ . When  $n$  is large, the dominant cost lies not in the abstract existence theory, but in the repeated dense linear algebra required to solve or approximate the system.

## 3.1 Sketch-and-Solve Principle

The basic randomized idea is to replace the original problem by a smaller one. Given a sketching matrix  $S \in \mathbb{R}^{s \times n}$  with  $s \ll n$ , one solves

$$\min_{x \in \mathbb{R}^d} \|SAx - Sb\|_2.$$

If  $S$  approximately preserves the geometry of the subspace carrying the least-squares information, then the reduced problem can deliver a solution whose residual is close to optimal at a lower computational cost. In operator language, the sketch should act almost isometrically on the column space of  $A$ , or more cautiously on the augmented subspace associated with both  $A$  and  $b$ . This subspace-embedding viewpoint is one of the organizing principles of modern RandNLA [16] [15].

The importance of this formulation is that it separates two questions that are easy to conflate. The first is approximation: does the sketch preserve enough geometry for the reduced problem to represent the original one faithfully? The second is computation: can the sketch be formed and used quickly enough to justify replacing the classical solver? Early least-squares algorithms already showed that this tradeoff can be favorable in practice [17] [18].

## 3.2 Subspace Embeddings

The geometric engine behind sketching is the notion of an oblivious subspace embedding (OSE). The definition is simple but its consequences run deep.

**Definition 3.1** (Oblivious Subspace Embedding). Let  $\epsilon \in (0, 1)$  and  $\delta \in (0, 1)$ . A distribution  $\mathcal{S}$  over  $\mathbb{R}^{s \times n}$  is a  $(d, \epsilon, \delta)$ -oblivious subspace embedding if, for every fixed subspace  $\mathcal{V} \subseteq \mathbb{R}^n$  of dimension at most  $d$ ,

$$\Pr_{S \sim \mathcal{S}} \left[ \forall v \in \mathcal{V} : (1 - \epsilon) \|v\|_2^2 \leq \|Sv\|_2^2 \leq (1 + \epsilon) \|v\|_2^2 \right] \geq 1 - \delta.$$

The key point is that the sketching matrix is drawn independently of the subspace. In the least-squares context, the relevant subspace is the column space of  $A$  (or the augmented space spanned by the columns of  $[A \ b]$ ), and the OSE property guarantees that the geometry of that subspace is approximately preserved after compression.

The squared-norm convention is the cleanest one for later spectral statements, because it is equivalent to controlling the Gram operator  $U^\top S^\top S U$  on an orthonormal basis  $U$  of the embedded subspace. It immediately implies the unsquared estimate

$$\sqrt{1 - \epsilon} \|v\|_2 \leq \|Sv\|_2 \leq \sqrt{1 + \epsilon} \|v\|_2,$$

so no geometric content is lost by working with squares.

The OSE condition can be interpreted as a uniform Johnson–Lindenstrauss property on subspaces. This connection is made precise by the Johnson–Lindenstrauss lemma, which provides the foundational concentration result.

**Lemma 3.2** (Johnson–Lindenstrauss). Let  $\epsilon \in (0, 1)$  and let  $\mathcal{P}$  be a finite set of  $p$  points in  $\mathbb{R}^n$ . If  $S \in \mathbb{R}^{s \times n}$  has entries drawn independently from  $\mathcal{N}(0, 1/s)$  and

$$s \geq \frac{4 \log p}{\epsilon^2/2 - \epsilon^3/3},$$

then with probability at least  $1 - 1/p$ ,

$$(1 - \epsilon) \|u - v\|_2^2 \leq \|S(u - v)\|_2^2 \leq (1 + \epsilon) \|u - v\|_2^2$$

for every pair  $u, v \in \mathcal{P}$ .

The JL lemma is a statement about finite point sets, but the extension to continuous subspaces follows by a net argument. A  $\gamma$ -net of the unit ball in a  $d$ -dimensional subspace has cardinality at most  $(3/\gamma)^d$ ; applying JL to this net and using the Lipschitz continuity of the sketching operator yields the subspace embedding property. The net argument is the standard bridge from pointwise concentration to uniform geometric control.

### 3.3 Gaussian Sketches

The first sketch family considered in this thesis is the Gaussian sketch. Its entries are independent normal random variables with variance  $1/s$ , so that

$$S_{ij} \sim \mathcal{N}(0, 1/s).$$

This scaling is chosen so that the sketch is isotropic in expectation. In particular, for a fixed vector  $x \in \mathbb{R}^n$ , one has the identity

$$\mathbb{E}\|Sx\|_2^2 = \|x\|_2^2,$$

which explains why Gaussian sketches are a natural baseline in both theory and computation.

The main theoretical advantage of the Gaussian model is that the concentration of quadratic forms under Gaussian matrices is well understood. The following result is the central embedding guarantee.

**Proposition 3.3 (Gaussian OSE).** Let  $S \in \mathbb{R}^{s \times n}$  be a Gaussian sketch with entries  $S_{ij} \sim \mathcal{N}(0, 1/s)$ . For any fixed matrix  $U \in \mathbb{R}^{n \times d}$  with orthonormal columns and any  $\epsilon \in (0, 1)$ ,

$$\Pr\left[\|U^\top S^\top S U - I_d\|_2 > \epsilon\right] \leq 2d \cdot \exp\left(-\frac{s(\epsilon^2 - \epsilon^3)}{4}\right).$$

In particular, if  $s \geq 4(d \log d + \log(2/\delta))/\epsilon^2$ , then  $S$  is a  $(d, \epsilon, \delta)$ -OSE.

**Proof (Proof sketch).** For a fixed unit vector  $u \in \mathbb{R}^n$ , the random variable  $\|Su\|_2^2$  is a sum of  $s$  independent  $\chi_1^2/s$  variables. By standard concentration inequalities for chi-squared random variables (the Laurent–Massart bound),

$$\Pr\left[\left|\|Su\|_2^2 - 1\right| > \epsilon\right] \leq 2 \exp\left(-\frac{s(\epsilon^2 - \epsilon^3)}{4}\right).$$

To pass from a single direction to the entire subspace, cover the unit ball of  $\text{col}(U)$  by a  $\gamma$ -net of cardinality at most  $(3/\gamma)^d$  with  $\gamma = \epsilon/2$ . A union bound over the net, combined with the Lipschitz property of the operator norm, yields the stated inequality. The dimension requirement follows by setting the failure probability to  $\delta$  and solving for  $s$ .  $\square$

The logarithmic dependence on the ambient dimension  $n$  is absent: the embedding quality depends only on the intrinsic dimension  $d$  and the sketch size  $s$ . This dimension-free character is the theoretical reason why Gaussian sketches are powerful regardless of how large the original problem is.

Closely related are dense sign sketches, often called Rademacher sketches, whose entries are independent  $\pm 1/\sqrt{s}$ . They retain much of the same isotropic behavior while replacing Gaussian sampling by a simpler discrete distribution. The OSE guarantee for Rademacher sketches is essentially identical to [Proposition 3.3  \$\rightarrow\$  p.13](#), with the same sketch-size requirement up to constants; the proof uses the Hanson–Wright inequality instead of chi-squared concentration.

### 3.4 The Subsampled Randomized Hadamard Transform

The third family relevant here is the subsampled randomized Hadamard transform (SRHT). Structured sketches of this type combine random signs, a fast orthogonal transform, and subsampling. They are attractive because they preserve much of the embedding behavior of dense sketches while allowing substantially faster sketch formation when the data layout permits fast transforms.

The SRHT is defined as follows. Let  $n = 2^p$  for some integer  $p$ . Denote by  $H_n \in \mathbb{R}^{n \times n}$  the normalized Walsh–Hadamard matrix (satisfying  $H_n H_n^\top = I_n$ ), and let  $D \in \mathbb{R}^{n \times n}$  be a diagonal matrix whose diagonal entries are independent uniform random signs. Let  $R \in \mathbb{R}^{s \times n}$  be a matrix that selects  $s$  rows uniformly at random without replacement. The SRHT sketch is then

$$S = \sqrt{\frac{n}{s}} R H_n D.$$

The Hadamard matrix can be applied in  $O(n \log n)$  operations via the fast Walsh–Hadamard transform, which makes the total cost of forming  $SA$  equal to  $O(nd \log n)$  instead of the  $O(nds)$  cost of a dense Gaussian sketch. The tradeoff is that the SRHT introduces additional combinatorial structure that makes the analysis more delicate.

With this normalization the SRHT is isotropic in expectation after random row selection, so that  $\mathbb{E}\|Sx\|_2^2 = \|x\|_2^2$  for fixed  $x$ . This is the structured analogue of the Gaussian scaling discussed earlier.

**Proposition 3.4 (SRHT Embedding).** Let  $S \in \mathbb{R}^{s \times n}$  be an SRHT sketch. For any fixed matrix  $U \in \mathbb{R}^{n \times d}$  with orthonormal columns and any  $\epsilon \in (0, 1)$ ,

$$\Pr\left[\|U^\top S^\top S U - I_d\|_2 > \epsilon\right] \leq \delta$$

provided  $s \geq C \frac{d \log n}{\epsilon^2} \log\left(\frac{d \log n}{\epsilon^2 \delta}\right)$  for a universal constant  $C$ .

The dependence on  $\log n$  is the price of the structure: the SRHT needs a slightly larger sketch than the Gaussian model to achieve the same embedding quality, but the gain in sketch-formation time can be substantial. For this thesis, the contrast between Gaussian and SRHT sketches is essential. Gaussian sketches are the cleanest theoretical baseline, and SRHT-type sketches represent the first genuinely structured option whose computational appeal may become decisive for large problems. The SRHT also plays a methodological role: because it is deterministic once the random signs and row indices are fixed, it is easier to reason about in a mixed-precision setting where the randomization and the arithmetic perturbations should be kept conceptually separate.

### 3.5 Residual and Solution Guarantees

The subspace-embedding property leads directly to approximation guarantees for the sketched least-squares problem. The following result is the basic accuracy statement for the sketch-and-solve paradigm.

**Proposition 3.5** (Sketched LS Residual Bound). Let  $A \in \mathbb{R}^{n \times d}$  have full column rank, let  $x_\star = \arg \min \|Ax - b\|_2$ , and let  $\hat{x} = \arg \min \|SAx - Sb\|_2$ . If  $S$  is a  $(d + 1, \epsilon, \delta)$ -OSE with  $\epsilon < 1$ , then with probability at least  $1 - \delta$ ,

$$\|A\hat{x} - b\|_2^2 \leq \frac{1 + \epsilon}{1 - \epsilon} \|Ax_\star - b\|_2^2,$$

and hence

$$\|A\hat{x} - b\|_2 \leq \sqrt{\frac{1 + \epsilon}{1 - \epsilon}} \|Ax_\star - b\|_2.$$

**Proof** (Proof sketch). Decompose  $b = Ax_\star + r_\star$  where  $r_\star$  is the optimal residual. Write  $\hat{x} = x_\star + h$ . The embedding property on the subspace  $\text{col}([A \ r_\star])$  gives

$$(1 - \epsilon) \|Ah + r_\star\|_2^2 \leq \|S(Ah + r_\star)\|_2^2 \leq (1 + \epsilon) \|Ah + r_\star\|_2^2.$$

Since  $\hat{x}$  minimizes  $\|S(Ax - b)\|_2$ , the optimality of  $\hat{x}$  for the sketched problem implies

$$\|SA\hat{x} - Sb\|_2^2 \leq \|SAx_\star - Sb\|_2^2 \leq (1 + \epsilon) \|r_\star\|_2^2.$$

Combining the two inequalities and solving for  $\|A\hat{x} - b\|_2$  yields the stated bound.  $\square$

The residual guarantee is natural but not the only quantity of interest. In many applications, one also cares about the solution error  $\|\hat{x} - x_\star\|_2$ . A bound on the solution error requires additional control over the geometry of the problem, specifically the extent to which the residual contributes to the solution perturbation.

**Remark 3.6** (Solution error versus residual error). Residual guarantees do not automatically imply a comparably small perturbation of the minimizer. In the inconsistent case, the sensitivity of  $x_\star$  is additionally filtered through the conditioning of  $A$  and by the size of the optimal residual relative to the solution itself. This is why the numerical experiments track both residual ratio and relative solution error. The residual bound is the robust first statement, while a formal solution-error theorem requires more careful assumptions than are imposed here.  $\sqcup$

These observations together explain the experimental expectations: residual quality stabilizes quickly as  $s$  grows, while solution error depends on both  $\epsilon$  and  $\kappa_2(A)$ . The transition from residual-dominated to solution-error-dominated behavior is one of the main features that the numerical experiments are designed to capture.

### 3.6 Conditioning of the Sketched System

Even when the sketch provides a good embedding in exact arithmetic, the numerical behavior of the reduced problem depends critically on the conditioning of  $SA$ . The following result connects the embedding quality to the spectral properties of the compressed operator.

**Proposition 3.7** (Conditioning of the Sketched Matrix). Let  $A \in \mathbb{R}^{n \times d}$  have full column rank with singular values  $\sigma_1 \geq \dots \geq \sigma_d > 0$ , and let  $S$  be a  $(d, \epsilon, \delta)$ -OSE. Then with probability at least  $1 - \delta$ ,

$$\sqrt{1 - \epsilon} \sigma_d \leq \sigma_{\min}(SA) \leq \sigma_{\max}(SA) \leq \sqrt{1 + \epsilon} \sigma_1,$$

and therefore

$$\kappa_2(SA) \leq \sqrt{\frac{1 + \epsilon}{1 - \epsilon}} \kappa_2(A).$$

This is a direct consequence of the embedding property applied to the left and right singular vectors of  $A$ . The important qualitative message is that a good subspace embedding cannot dramatically worsen the condition number: if  $\epsilon$  is small, the sketched system is only slightly less well-conditioned than the original.

However, the proposition assumes exact arithmetic. In finite precision, the situation is more nuanced. The computed sketch  $\text{fl}(SA)$  is a perturbation of the ideal  $SA$ , and the effective condition number of the computed system may be larger than  $\kappa_2(SA)$  by an amount that depends on the precision of the sketch-forming arithmetic and on the dimensions  $n$  and  $s$ . This is precisely where the interaction between randomized approximation and floating-point perturbation becomes important, and it motivates the mixed-precision analysis in [Chapter 2  \$\rightarrow\$  p.6](#).

In practice, the condition number of  $SA$  is a useful diagnostic. If  $\kappa_2(SA)$  remains bounded as  $s$  varies, the sketch is preserving the spectral structure of the original problem; if it grows sharply, the sketch dimension may be insufficient, or the sketch may be failing to embed the relevant subspace. The numerical experiments in [Chapter 5  \$\rightarrow\$  p.29](#) track  $\kappa_2(SA)$  as a function of  $s$  for exactly this reason.

### 3.7 From Embedding to Solver

The sketching matrix is only one part of the algorithm. Once the reduced operator  $SA$  has been formed, one still has to solve the compressed least-squares problem in a numerically sensible way. This is where conditioning re-enters the story. A good embedding should preserve not only residual geometry but also enough spectral structure that the reduced problem remains well behaved.

In practical randomized least-squares solvers, two related paradigms appear repeatedly. The first is sketch-and-solve, in which one directly solves the compressed

problem. The second is sketch-and-precondition, in which the sketch is used to build a preconditioner for a more accurate iterative method or refinement procedure. Blendenpik is one of the landmark examples: it combines randomized mixing with a high-quality dense least-squares solve and shows that randomized preprocessing can improve performance without abandoning numerical linear algebra discipline [18]. More recent work goes further and shows that randomized least-squares solvers can be designed to be forward stable, and in some formulations even backward stable, rather than merely fast on average [19] [20].

This stability question is especially important for the present thesis. It is not enough to know that a sketch preserves the least-squares objective in exact arithmetic. One also needs to know how the conditioning of  $SA$ , and later of any preconditioned or refined system, interacts with the finite-precision arithmetic discussed in [Chapter 2<sup>→p.6</sup>]. In that sense, modern RandNLA has moved beyond the initial question of whether randomization can accelerate least squares; it now asks under what conditions randomized solvers can be trusted as serious numerical methods [21] [15].

### 3.8 Sketch-and-Precondition

The sketch-and-solve viewpoint is conceptually clean, but it is not always the best numerical endpoint. A second paradigm uses the sketch to build a preconditioner for the original problem rather than treating the reduced problem as the final approximation. In broad terms, one computes a factorization of the compressed matrix  $SA$ , uses that factorization to define a right preconditioner  $R^{-1}$ , and then solves the transformed problem

$$\min_y \|AR^{-1}y - b\|_2, \quad x = R^{-1}y,$$

typically by an iterative method or by refinement around a high-quality residual computation.

This shift in viewpoint matters because the sketch no longer bears the entire burden of solution accuracy. Its role is to improve the conditioning of the operator seen by the downstream solver. If  $SA$  captures the geometry of the column space well enough, then the preconditioned matrix  $AR^{-1}$  can become substantially better behaved than the original least-squares system. Blendenpik is the classical example of this philosophy: randomized mixing and sampling are used to construct a preconditioner, but the final method remains firmly tied to stable least-squares solving rather than replacing it by a one-shot compressed surrogate [18].

At the same time, recent analysis shows that the numerical details of this paradigm matter more than one might expect. Meier, Nakatsukasa, Townsend, and Webb show that the commonly used form of sketch-and-precondition can lose numerical stability on ill-conditioned problems, even though the underlying sketch may be geometrically accurate in exact arithmetic [22]. Their analysis makes a crucial

distinction between having a good randomized preconditioner in principle and embedding that preconditioner into a floating-point algorithm in a stable way. In particular, they advocate solving the preconditioned least-squares problem with an unpreconditioned iterative method applied to the transformed system, rather than relying on the numerically fragile form of the standard algorithm.

This observation sharpens the role of conditioning in the thesis. The object of interest is not merely whether  $\kappa_2(SA)$  is moderate, but whether the entire solver architecture built around  $SA$  respects that favorable conditioning once rounding errors enter. It is precisely this distinction that motivates the later mixed-precision experiments. A lower-precision sketching stage may be acceptable if the resulting preconditioner still places the downstream method in a benign regime; it is unacceptable if the arithmetic perturbation destroys the conditioning improvement that the randomized step was supposed to provide.

For the present thesis, this paradigm is important for two reasons. First, it clarifies that randomized least squares is not exhausted by direct sketch-and-solve methods; preconditioning is equally central to the modern literature. Second, it provides a natural bridge to mixed precision. Recent work by Carson and Daužickaitė studies precisely this issue for sketching-based preconditioners combined with GMRES-based iterative refinement, showing how separate precision choices in the sketching and factorization stages affect whether the preconditioner remains effective [11]. This is exactly the kind of coupled approximation-and-arithmetic question that later experiments should address.

### 3.9 A First Experiment

The initial computational study in this thesis is deliberately simple. A random matrix  $A \in \mathbb{R}^{n \times d}$  is generated together with a reference solution  $x_*$ , and the right-hand side is taken as either

$$b = Ax_*$$

in the consistent case or

$$b = Ax_* + e$$

in the noisy case, where  $e$  is a perturbation vector. The current Octave prototype uses Gaussian sketches, while the R package extends the same benchmark structure to Rademacher sketches and the SRHT. For each sketch size  $s$ , several independent sketches are drawn, the sketched least-squares problems are solved, and the following quantities are recorded:

- the mean absolute error of the averaged solution,
- the residual ratio relative to the unsketched least-squares solution,
- the relative solution error with respect to a reference solve,
- the condition number of the sketched matrix  $SA$ .

This experiment plays an important organizational role. Mathematically, it gives a concrete setting in which subspace-embedding heuristics can be compared against observed residual and solution behavior. Computationally, it provides a common benchmark across the current prototype implementations, which already cover Gaussian, Rademacher, and SRHT sketches. In that sense, it is the smallest setting in this project where geometric approximation, conditioning of the compressed operator, and measured computation are meant to line up exactly.

### 3.10 Why This Experiment Matters

Even before formal guarantees are introduced, the experiment highlights the main questions that drive the later chapters. How large must the sketch dimension  $s$  be before the residual becomes reliably close to optimal? How does the conditioning of the compressed operator  $SA$  change as  $s$  varies? Which sketch family gives the best compromise between simplicity, structure, and numerical behavior? Once mixed precision is introduced, the same framework also makes it possible to ask which arithmetic stages tolerate lower precision and which do not.

The main topics to be developed further are therefore clear: subspace embeddings as the geometric core of sketching, Gaussian and structured sketches as competing algorithmic choices, residual and solution guarantees for sketch-and-solve methods, and the conditioning and finite-precision behavior of the reduced systems. Those topics place the present thesis within the broader RandNLA literature while keeping the attention on the concrete least-squares problems that the code and experiments in this repository already target.

At the same time, least squares is not the final mathematical endpoint of the thesis. It is the first model problem in which sketching, conditioning, and finite precision interact transparently. In the later covariance-transport chapter, the same perturbation viewpoint is transferred from sketched least-squares operators to empirical covariance operators and the matrix functions built from them. The role of the present chapter is therefore foundational: it supplies the geometric and numerical language that the dependence-correction analysis will reuse in a more application-aware setting.

# 4 Randomized Covariance Transport

This chapter formulates the climate-facing problem in the thesis that is closest to its numerical core. The climate-model application matters, but the main mathematical object is not the full post-processing workflow. It is a regularized dependence-correction map built from empirical covariance operators after marginal adjustment. This viewpoint keeps the discussion anchored in randomized numerical linear algebra and mixed-precision perturbation analysis while still preserving a clear connection to multivariate bias correction.

Existing multivariate bias-correction methods already show why dependence structure matters. Quantile-delta mapping (QDM) corrects marginal distributions while preserving modeled changes in quantiles [23]. Methods such as MBCp and MBCr combine marginal correction with dependence adjustment, and MBCn and later rank-based approaches push further toward correcting the full multivariate distribution [24] [25] [26] [27]. At the same time, the climate literature repeatedly warns that bias correction should not be treated as a black-box distributional repair, especially when the method modifies structures that matter dynamically or physically [28]. For the present thesis, this leads to a narrower and more analyzable question: can one replace an expensive dependence-correction step by a randomized low-rank approximation whose finite-precision behavior can still be controlled?

The main claim of this chapter is therefore methodological. Rather than analyze the full nonlinear climate workflow at once, we isolate a covariance-based transport problem in a latent dependence space. This reduction is still rich enough to capture the main tradeoffs between sketch dimension, regularization, conditioning, and arithmetic precision, but it is much cleaner than attempting a direct perturbation analysis of a fully nonlinear multivariate adjustment scheme.

## 4.1 Latent Dependence Representation

Let

$$X_h \in \mathbb{R}^{n \times p}, \quad X_f \in \mathbb{R}^{m \times p}, \quad Y_h \in \mathbb{R}^{n \times p}$$

denote historical climate-model data, future climate-model data, and historical reference data, respectively. The column dimension  $p$  may encode several physical variables, nearby locations, lags, seasonal features, or any local multivariate state used for downstream post-processing.

The first stage remains standard. Each marginal series is corrected by a univariate method of QDM type,

$$\widetilde{X}_h, \widetilde{X}_f = \text{QDM}(X_h, X_f; Y_h),$$

so that the eventual output inherits the desired marginal behavior from the reference historical data together with modeled changes in quantiles [23]. What remains is the dependence correction.

To express dependence in a form closer to linear algebra, it is convenient to pass to a latent Gaussianized space. The precise standardization map can be chosen in several closely related ways, but it should avoid empirical probabilities equal to 0 or 1, since  $\Phi^{-1}$  would then produce infinities. A convenient finite definition is obtained from plotting-position ranks. For each column  $j$ , write

$$u_{X,ij} = \frac{\text{rank}(\widetilde{X}_h(i, j)) - \frac{1}{2}}{n}, \quad u_{Y,ij} = \frac{\text{rank}(Y_h(i, j)) - \frac{1}{2}}{n},$$

and similarly

$$u_{f,ij} = \frac{\text{rank}(\widetilde{X}_f(i, j)) - \frac{1}{2}}{m}$$

for the future sample. Ties are handled by average ranks; any fixed deterministic tie-breaking rule would serve equally well. We then define

$$Z_X(i, j) = \Phi^{-1}(u_{X,ij}), \quad Z_Y(i, j) = \Phi^{-1}(u_{Y,ij}),$$

and similarly obtain  $Z_f$  from  $u_f$ . In this representation, marginal correction is separated from dependence correction: the marginals are handled before and after the latent transform, while the multivariate adjustment acts on the standardized latent variables.

This reduction is not meant to claim that all relevant climate dependence is Gaussian. It is a modeling step that isolates a dependence operator that is computationally tractable and mathematically interpretable. In particular, it allows the dependence-correction stage to be expressed through covariance operators in a finite-dimensional inner-product space.

## 4.2 Regularized Covariance Transport

Assume that the columns of  $Z_X$  and  $Z_Y$  have been centered. Their empirical covariance operators are

$$C_X = \frac{1}{n} Z_X^\top Z_X, \quad C_Y = \frac{1}{n} Z_Y^\top Z_Y.$$

The dependence-correction step is then represented by the regularized transport map

$$T_\lambda = (C_X + \lambda I)^{-1/2} (C_Y + \lambda I)^{1/2},$$

where  $\lambda > 0$  is a regularization parameter.

The role of  $\lambda$  is both numerical and conceptual. Numerically, it keeps the source and target operators strictly positive definite and prevents the inverse square root from reacting catastrophically to small singular values or nearly redundant

variables. Conceptually, it acknowledges that in high-dimensional post-processing tasks one should not insist on recovering dependence structure below the scale at which the empirical covariance is already poorly determined.

The corrected latent future data are then obtained as

$$Z_f^{\text{corr}} = Z_f T_\lambda.$$

**Remark 4.1** (Regularization bias). For  $\lambda = 0$ , the idealized transport map  $T = C_X^{-1/2} C_Y^{1/2}$  satisfies

$$T^\top C_X T = C_Y$$

whenever the inverse square root is well defined. For  $\lambda > 0$ , the regularized map instead satisfies

$$T_\lambda^\top (C_X + \lambda I) T_\lambda = C_Y + \lambda I,$$

and therefore transports the regularized covariance exactly, not the original unregularized covariance. Thus  $\lambda$  controls both numerical stability and modelling bias.  $\lrcorner$

Finally, because a linear map in latent space will generally disturb the exact marginal distributions, the latent correction is used only to update ranks. For each variable, the values from the QDM-corrected future sample  $\widetilde{X}_f(:, j)$  are sorted and then rearranged according to the ranks of  $Z_f^{\text{corr}}(:, j)$ . The final output therefore combines

marginals from QDM + dependence from covariance transport.

This construction should be viewed as a lower-rank and more analyzable alternative to full multivariate distribution matching. It is less ambitious than MBCn in what it tries to capture, but it is correspondingly better aligned with the perturbation tools of numerical linear algebra.

### 4.3 Randomized Low-Rank Approximation

The direct formation and decomposition of  $C_X$  and  $C_Y$  is unattractive when  $p$  is large. The thesis therefore proposes to approximate the covariance operators by randomized low-rank surrogates. Let  $\Omega \in \mathbb{R}^{p \times \ell}$  be a sketching matrix with  $\ell = r + q$ , where  $r$  is the target rank and  $q$  an oversampling parameter. Then the dominant range of  $C_X$  may be sampled through

$$Y_X = C_X \Omega = \frac{1}{n} Z_X^\top (Z_X \Omega),$$

and analogously for  $C_Y$ . This is precisely the type of factorization-friendly structure for which randomized low-rank approximation is effective [14] [15].

If  $Q_X$  is an orthonormal basis for the range of  $Y_X$ , the covariance operator is approximated by

$$C_X \approx Q_X B_X Q_X^\top, \quad B_X = Q_X^\top C_X Q_X,$$

with an analogous approximation for  $C_Y$ . The transport map is then built from

$$\hat{C}_{X,\lambda} = Q_X B_X Q_X^\top + \lambda I, \quad \hat{C}_{Y,\lambda} = Q_Y B_Y Q_Y^\top + \lambda I.$$

The computational advantage is immediate: the numerically demanding work is concentrated in matrix products of the form  $Z\Omega$  and  $Z^\top(Z\Omega)$ , while the eigenanalysis is transferred to the small dense matrices  $B_X$  and  $B_Y$ .

**Proposition 4.2** (Randomized covariance approximation). Let  $C \in \mathbb{R}^{p \times p}$  be symmetric positive semidefinite, and let  $Q$  be produced by a randomized range finder with target rank  $r$  and oversampling parameter  $q$ . Then the compression error

$$\|(I - QQ^\top)C\|_2$$

is controlled by the spectral tail of  $C$  beyond rank  $r$ . In particular, standard range-finder bounds imply that there exist oversampling-dependent constants  $c_{r,q}$  and  $d_{r,q}$  such that

$$\mathbb{E} \|(I - QQ^\top)C\|_2 \leq c_{r,q} \mu_{r+1}(C) + d_{r,q} \left( \sum_{j>r} \mu_j(C)^2 \right)^{1/2},$$

where  $\mu_1(C) \geq \mu_2(C) \geq \dots \geq 0$  are the eigenvalues of  $C$  in nonincreasing order [Chs. 10–11 [14], Sec. 8 [15]].

**Proof** (Proof sketch). This is the standard randomized range-finder estimate specialized to a symmetric positive semidefinite matrix. Since  $C$  is self-adjoint, its singular values are precisely the eigenvalues  $\mu_j(C)$ .  $\square$

Moreover, the regularized matrix functions of a low-rank surrogate can be applied without forming a full dense  $p \times p$  matrix. If  $Q \in \mathbb{R}^{p \times r}$  has orthonormal columns and  $B \in \mathbb{R}^{r \times r}$  is symmetric positive semidefinite, then

$$(QBQ^\top + \lambda I)^{-1/2} = \lambda^{-1/2} I + Q \left( (B + \lambda I_r)^{-1/2} - \lambda^{-1/2} I_r \right) Q^\top,$$

and similarly

$$(QBQ^\top + \lambda I)^{1/2} = \lambda^{1/2} I + Q \left( (B + \lambda I_r)^{1/2} - \lambda^{1/2} I_r \right) Q^\top.$$

These identities show that the transport layer only needs matrix functions of the small core matrix  $B + \lambda I_r$ , together with applications of  $Q$  and  $Q^\top$ .

The point is not only efficiency. A low-rank approximation introduces an explicit randomized error scale. Once this scale is visible, it becomes meaningful to ask whether lower precision is acceptable because its rounding error remains smaller than the approximation error already introduced by the randomized compression.

## 4.4 Mixed Precision and Balancing Inexactness

This is the place where the climate application and the mixed-precision chapter meet most naturally. The important question is not whether lower precision is faster in the abstract. It is whether reduced precision changes the dependence correction less than the inexactness already present in the modeling pipeline. This is exactly the viewpoint advocated in recent mixed-precision work on balancing multiple sources of inexactness [13] [2].

For the proposed method, the natural arithmetic split is the following. The bulk products

$$Z\Omega, \quad Z^\top(Z\Omega)$$

are the most plausible candidates for reduced precision. By contrast, the orthogonalization that produces  $Q$ , the small eigendecompositions of  $B_X$  and  $B_Y$ , and the evaluation of matrix square roots and inverse square roots are more sensitive and are therefore better kept in double precision. This follows the general mixed-precision pattern already visible in least-squares sketching and refinement methods: use cheaper arithmetic where the computation is large and structurally robust, and keep higher precision where instability would be amplified by conditioning [11] [5].

Accordingly, the computed covariance approximation should be written as

$$\hat{C}_X = C_X + E_{\text{rand},X} + E_{\text{fp},X}, \quad \hat{C}_Y = C_Y + E_{\text{rand},Y} + E_{\text{fp},Y},$$

where  $E_{\text{rand}}$  denotes randomized approximation error and  $E_{\text{fp}}$  floating-point error. A concrete floating-point model enters already at the sample matrix level. If the dominant product is formed in lower precision with unit roundoff  $u_\ell$ , then one may write

$$\hat{Y}_X = \text{fl}_{u_\ell} \left( \frac{1}{n} Z_X^\top (Z_X \Omega) \right) = C_X \Omega + \Delta Y_X,$$

and analogously for  $\hat{Y}_Y$ . Under the standard matrix-product rounding model,

$$\|\Delta Y_X\|_2 \lesssim \gamma_n^{(u_\ell)} \frac{\|Z_X\|_2^2}{n} \|\Omega\|_2, \quad \gamma_n^{(u_\ell)} = \frac{nu_\ell}{1 - nu_\ell},$$

with the same form for  $\Delta Y_Y$  [2]. Thus the floating-point contribution depends not only on the unit roundoff, but also on the accumulation length, the scale of the latent data, the norm of the sketching operator, and on whether accumulation is performed in low or high precision. The key regime is the balanced one,

$$\|E_{\text{fp}}\| \lesssim \|E_{\text{rand}}\|,$$

or more generally when the floating-point perturbation remains below the statistical or modeling tolerance of the application. In that regime, computing the sketch in full double precision would amount to solving one part of the numerical problem much more accurately than the rest of the pipeline warrants.

## 4.5 Perturbation Perspective

The central analytical task is not to study the entire bias-correction workflow at once, but to bound the perturbation of the transport map induced by the covariance approximation. Define

$$A = C_X + \lambda I, \quad B = C_Y + \lambda I,$$

and let

$$\hat{A} = A + \Delta A, \quad \hat{B} = B + \Delta B$$

be the regularized approximations produced by randomized low-rank compression and finite-precision arithmetic. Then

$$T_\lambda = A^{-1/2} B^{1/2}, \quad \hat{T}_\lambda = \hat{A}^{-1/2} \hat{B}^{1/2}.$$

The natural forward quantities are

$$\|T_\lambda - \hat{T}_\lambda\|_2 \quad \text{and} \quad \|Z_f(T_\lambda - \hat{T}_\lambda)\|_F.$$

At a first formal level one may split the transport error as

$$\|T_\lambda - \hat{T}_\lambda\|_2 \leq \|A^{-1/2}\|_2 \|B^{1/2} - \hat{B}^{1/2}\|_2 + \|A^{-1/2} - \hat{A}^{-1/2}\|_2 \|\hat{B}^{1/2}\|_2.$$

This identity already reveals the main conditioning mechanism. Small covariance perturbations are not enough by themselves. Their effect is filtered through matrix square roots and, more severely, inverse square roots. Therefore the decisive quantity is not only the size of  $\Delta A$  and  $\Delta B$ , but also the spectral safety margin created by the regularization parameter  $\lambda$ .

If

$$\|\Delta A\|_2 \leq \eta_A, \quad \|\Delta B\|_2 \leq \eta_B,$$

the later perturbation analysis can be organized into a small chain of deterministic statements.

**Proposition 4.3** (Positive definiteness under perturbation). Let  $A = C_X + \lambda I$  and  $\hat{A} = A + \Delta A$ , where  $A$  and  $\hat{A}$  are symmetric. If  $\|\Delta A\|_2 < \lambda_{\min}(A)$ , then  $\hat{A}$  remains symmetric positive definite and

$$\lambda_{\min}(\hat{A}) \geq \lambda_{\min}(A) - \|\Delta A\|_2.$$

The same statement holds for  $B$  and  $\hat{B}$ .

**Proof** (Proof sketch). For any unit vector  $x$ ,

$$x^\top \hat{A} x = x^\top A x + x^\top \Delta A x \geq \lambda_{\min}(A) - \|\Delta A\|_2.$$

Taking the infimum over  $\|x\|_2 = 1$  gives the bound.  $\square$

**Proposition 4.4** (Perturbation of regularized square roots). Assume that  $A$ ,  $\hat{A}$ ,  $B$ , and  $\hat{B}$  are symmetric positive definite, and set

$$\alpha_0 = \min\{\lambda_{\min}(A), \lambda_{\min}(\hat{A})\}, \quad \beta_0 = \min\{\lambda_{\min}(B), \lambda_{\min}(\hat{B})\}.$$

Then

$$\|A^{-1/2} - \hat{A}^{-1/2}\|_2 \leq \frac{1}{2\alpha_0^{3/2}} \|\Delta A\|_2,$$

and

$$\|B^{1/2} - \hat{B}^{1/2}\|_2 \leq \frac{1}{2\beta_0^{1/2}} \|\Delta B\|_2.$$

**Proof** (Proof sketch). These inequalities are standard perturbation bounds for matrix functions on the positive definite cone. One route is to write the Fréchet derivative of the square-root map through a Sylvester equation and then apply a mean-value bound for matrix functions; see [Chs. 3 and 6 \[29\]](#). The inverse-square-root estimate then follows from the identity

$$A^{-1/2} - \hat{A}^{-1/2} = A^{-1/2}(\hat{A}^{1/2} - A^{1/2})\hat{A}^{-1/2},$$

combined with the square-root perturbation bound. The important qualitative point is the scaling: the inverse square root is more sensitive near small eigenvalues, with amplification proportional to  $\alpha_0^{-3/2}$ .  $\square$

**Theorem 4.5** (Perturbation of covariance transport). Assume that the hypotheses of the previous proposition hold. Then

$$\|T_\lambda - \hat{T}_\lambda\|_2 \leq \|A^{-1/2}\|_2 \|B^{1/2} - \hat{B}^{1/2}\|_2 + \|A^{-1/2} - \hat{A}^{-1/2}\|_2 \|\hat{B}^{1/2}\|_2,$$

and therefore

$$\|T_\lambda - \hat{T}_\lambda\|_2 \leq \frac{\|A^{-1/2}\|_2}{2\beta_0^{1/2}} \eta_B + \frac{\|\hat{B}^{1/2}\|_2}{2\alpha_0^{3/2}} \eta_A.$$

In particular, there exists an amplification factor  $K_\lambda(C_X, C_Y)$  such that

$$\|T_\lambda - \hat{T}_\lambda\|_2 \lesssim K_\lambda(C_X, C_Y)(\eta_A + \eta_B),$$

where  $K_\lambda$  deteriorates as the regularized spectra approach zero.

**Proof** (Proof sketch). Split the difference as

$$T_\lambda - \hat{T}_\lambda = A^{-1/2}(B^{1/2} - \hat{B}^{1/2}) + (A^{-1/2} - \hat{A}^{-1/2})\hat{B}^{1/2},$$

then apply the triangle inequality and the previous proposition. The final bound is first order in  $\eta_A$  and  $\eta_B$ , with constants determined by the regularized spectral gaps.  $\square$

The randomized and arithmetic contributions can then be separated as

$$\eta_A \leq \eta_{\text{rand},A} + \eta_{\text{fp},A}, \quad \eta_B \leq \eta_{\text{rand},B} + \eta_{\text{fp},B}.$$

For spectra with visible decay, the randomized term is governed by the tail beyond the chosen rank, up to oversampling constants from the range-finder bound above [14] [15]. The floating-point term, by contrast, follows the sketch-product model above and is controlled by the working precision, the accumulation length, the scale of the latent data, and the norm of the sketching operator. The thesis question is precisely how these two error sources should be balanced against each other.

**Corollary 4.6** (Perturbation of corrected future data). Under the hypotheses of the theorem,

$$\|Z_f T_\lambda - Z_f \hat{T}_\lambda\|_F \leq \|Z_f\|_F \|T_\lambda - \hat{T}_\lambda\|_2.$$

**Proof.** This is the submultiplicativity of the Frobenius norm with the operator norm on the right.  $\square$

## 4.6 Algorithmic Consequences

This formulation leads to a concrete algorithmic decomposition.

- ( ) Correct marginals by QDM.
- ( ) Transform the corrected calibration and future samples to a latent rank-normal space.
- ( ) Approximate the calibration covariance operators by randomized low-rank sketches.
- ( ) Build a regularized covariance transport map from the low-rank surrogates.
- ( ) Apply the map to the future latent data.
- ( ) Restore the target marginals by rank rearrangement.

This is the natural architecture for the software contribution as well. The implementation should separate a randomized covariance-approximation layer, a transport-map layer, and a climate-facing wrapper that handles marginal correction and rank restoration. Such a decomposition keeps the numerical core reusable while preserving a clear application entry point.

The local `randMBC` prototype now realizes this decomposition explicitly. It exposes a deterministic marginal-correction and rank-restoration wrapper around a randomized covariance-transport core, together with a synthetic benchmark that compares the full regularized transport baseline, selected reference methods from the vendored `MBC` package, and the current randomized low-rank implementation.

The first controlled runs are informative even though they are not yet favorable to the randomized method: on the present synthetic benchmark, the deterministic full transport and the MBC references remain markedly more accurate than the current low-rank randomized surrogate. This should not be read as a failure of the covariance-transport viewpoint itself. Rather, it identifies more precisely where the open problem lies, namely in the approximation regime where low-rank compression becomes accurate enough that transport-map perturbations no longer dominate the downstream dependence correction.

The same decomposition also determines the experimental program. The decisive sweeps are not arbitrary. They should vary the approximation rank, the regularization parameter, and the arithmetic precision while tracking covariance error, transport-map error, and the induced dependence error on the corrected future data. In particular, one should expect a regime where single precision is essentially indistinguishable from double because randomized truncation already dominates, and another regime where further rank increase reveals floating-point error because the approximation error has become too small to hide it.

## 4.7 Role in the Thesis

The purpose of this chapter is not to claim that covariance transport solves the whole problem of multivariate climate bias correction. It does not replace the need for application-side diagnostics, and it should not be presented as a universal substitute for richer multivariate schemes such as MBCn. Its role is narrower and more precise. It identifies a mathematically serious dependence-correction problem that is simultaneously

- ( ) motivated by climate-model post-processing,
- ( ) expressible in the language of randomized low-rank approximation,
- ( ) sensitive to conditioning and regularization,
- ( ) and naturally suited to mixed-precision analysis.

For the later parts of the thesis, this chapter therefore acts as a bridge. Upstream, it uses the perturbation viewpoint of [Chapter 2<sup>→P.6</sup>] and the randomized approximation ideas from [Chapter 3<sup>→P.11</sup>]. Downstream, it provides the specific method family that the implementation and the climate-oriented experiments should test. In this sense, randomized covariance transport is the point at which the thesis becomes simultaneously application-aware and still unmistakably about numerical linear algebra.

# 5 Numerical Experiments

This chapter describes the experimental framework used in the repository for randomized least-squares solvers and for the later covariance-transport problem. The goal is not only to illustrate the theory, but to test whether the geometric statements from [Chapter 3](#)<sup>→ p.11</sup> and the perturbation viewpoint from [Chapter 4](#)<sup>→ p.20</sup> remain visible in computed residuals, computed solutions, covariance approximations, and transport errors. The experiments also provide the first controlled setting in which the floating-point questions from [Chapter 2](#)<sup>→ p.6</sup> can be studied in the same benchmark.

At the present stage, the experiments are intentionally synthetic. This is useful. By generating the data matrix, covariance structure, and reference solution directly, we can separate the effect of the sketch from the complications of a downstream application. The experiments are therefore simple enough to interpret and still rich enough to show the interaction between embedding quality, conditioning, regularization, and arithmetic.

## 5.1 Experimental Objective

The experiments are naturally divided into two parts. Part I studies sketched least-squares solvers as the first model problem. Part II studies randomized covariance transport as the climate-facing operator problem introduced in [Chapter 4](#)<sup>→ p.20</sup>. The repository currently implements mainly the first part, but the second part already has a clear mathematical specification.

For the least-squares experiments, the main question is how the quality of a sketched solve changes as the sketch dimension varies and as the sketch family changes. The repository currently tracks four quantities:

- the mean absolute error of the averaged sketch solution,
- the residual ratio relative to a reference least-squares solve,
- the relative solution error with respect to the reference solution,
- the condition number of the sketched matrix  $SA$ .

These quantities reflect different numerical phenomena. The residual ratio measures how well the sketched problem preserves the least-squares objective. The solution error is more sensitive to conditioning and therefore detects failures that may not be visible at the residual level. The condition number of  $SA$  is not an error measure by itself, but it often explains why two sketches with similar residual behavior may differ strongly at the level of the computed solution.

The experiments are therefore not only comparisons of random projections. They

test how approximation quality, solver sensitivity, and arithmetic interact in one concrete computational pipeline.

## 5.2 Synthetic Least-Squares Model

The current implementations use one common synthetic model. A matrix  $A \in \mathbb{R}^{n \times d}$  is generated with independent standard normal entries, and the reference solution is chosen as

$$x_\star = \mathbf{1} \in \mathbb{R}^d.$$

The right-hand side is then formed as

$$b = Ax_\star + e,$$

where  $e$  is either zero or a Gaussian perturbation scaled by a prescribed noise level. In this way, both the consistent and noisy regimes are available within the same template.

For each generated instance, a reference least-squares solution  $x_{\text{ref}}$  is computed from the full system. In the R implementation this is done with `qr.solve`, while the Octave prototypes use the backslash solver. The reference residual

$$r_{\text{ref}} = Ax_{\text{ref}} - b$$

provides the baseline against which the sketched residuals are compared. This is important because each trial is measured against the corresponding unsketched least-squares solve.

The synthetic model is deliberately modest. It does not yet attempt to mimic the spectral structure of a climate-data operator or the coherence patterns of an application matrix. Its role is to isolate the basic mechanisms of randomized least squares under controlled conditions.

## 5.3 Implemented Sketch Families

The repository currently supports three sketch families in the main R package: Gaussian sketches, Rademacher sketches, and the SRHT. The Octave prototypes currently cover Gaussian sketches and the SRHT. This reflects the current state of the software rather than a conceptual distinction. The R package is the intended long-term implementation base, while the Octave scripts remain useful as compact prototypes.

The Gaussian sketch is the cleanest theoretical baseline. The Rademacher sketch replaces Gaussian entries by random signs and tests how much of the same behavior survives under a simpler discrete distribution. The SRHT introduces structure and therefore changes both the cost model and the embedding constants. It is

the first sketch family in the project for which it is natural to ask whether the additional algebraic structure is worth the analytical complexity.

There is one practical restriction that matters experimentally. The current SRHT implementations require the row dimension to be a power of two. Therefore, the R and Octave code round  $n$  upward when needed before constructing the Hadamard-based sketch. This is a simple example of a theoretical assumption becoming an implementation condition.

## 5.4 Per-Trial Computation

For a fixed sketch size  $s$ , one trial proceeds as follows. A sketching matrix  $S \in \mathbb{R}^{s \times n}$  is drawn, the compressed system

$$SAx \approx Sb$$

is formed, and the reduced least-squares problem is solved. The computed solution for that trial is denoted by  $\hat{x}$ . The experiment then records:

$$\begin{aligned} \text{residual ratio} &= \frac{\|A\hat{x} - b\|_2}{\|Ax_{\text{ref}} - b\|_2}, \\ \text{solution error} &= \frac{\|\hat{x} - x_{\text{ref}}\|_2}{\|x_{\text{ref}}\|_2}, \\ \text{condition number} &= \kappa_2(SA). \end{aligned}$$

These definitions are aligned with the theoretical discussion in [Chapter 3<sup>→p.11</sup>](#). The residual ratio corresponds to the sketch-and-solve guarantee that the compressed problem should preserve the least-squares objective up to a controlled distortion. The solution error reflects the fact that preserving the residual does not automatically guarantee an equally small perturbation in the minimizer. The condition number of  $SA$  is recorded because it is one of the main mechanisms that links these two behaviors.

In addition, the implementations average the independent trials at each sketch size. If  $\hat{x}^{(1)}, \dots, \hat{x}^{(m)}$  are the trial solutions, the code forms their mean and reports the mean absolute error of this averaged estimator relative to the planted vector  $x_\star$ . This is not a standard RandNLA performance criterion, but it is useful here because it exposes the variability of the individual trial solutions.

## 5.5 Parameter Sweeps

The experiments vary the sketch dimension  $s$  over a prescribed range and repeat the randomized solve several times for each value. In the current R code, the generic routine `sketch_lstsq_experiment` accepts the parameters  $n$ ,  $d$ , a vector of sketch sizes, a trial count, a noise level, and the sketch family. Specialized

wrappers currently expose Gaussian and SRHT runs directly, but the underlying routine already supports all three sketch families.

The Octave scripts implement the same logic in a more direct form. The Gaussian prototype sweeps over `s_vals = 1:99` by default for a problem of size  $n = 100$ ,  $d = 10$ , while the SRHT prototype uses a coarser sweep adapted to the power-of-two requirement. In both cases, several independent sketches are drawn for each  $s$ , and the recorded history stores the sketch size together with the averaged error metrics.

This repeated-trial design is essential. A single randomized sketch may be atypical, especially for small sketch sizes. Averaging over several trials does not remove randomness from the method, but it makes the observed trends much more interpretable.

## 5.6 Precision-Sensitivity Baseline

The mixed-precision part of the repository is still at an early stage, but the current Octave Gaussian experiment already contains a useful first comparison. It can run either in ordinary double precision or, when the symbolic package is available, in a higher-precision mode used as a surrogate for reduced rounding error. This is not yet a full mixed-precision implementation in the sense discussed in [Chapter 2](#)<sup>→ p.6</sup>. It is a controlled way to ask whether a visible loss of accuracy is mainly geometric or mainly arithmetic.

That distinction matters. If increasing arithmetic precision leaves the curves essentially unchanged, the dominant limitation is likely the sketch itself or the conditioning of the reduced problem. If the higher-precision run materially improves the behavior, then floating-point perturbations are already visible at this synthetic scale. Therefore, the current precision experiment should be viewed as a diagnostic baseline for the later mixed-precision work.

## 5.7 How the Results Should Be Read

The main qualitative expectation is straightforward. As the sketch dimension grows, the residual ratio should approach one, because the compressed problem should preserve the geometry of the original least-squares problem more and more accurately. The solution error should also generally improve, but often less monotonically and sometimes more slowly, because it depends more strongly on the conditioning of the sketched operator. Therefore, the condition number  $\kappa_2(SA)$  should be read together with the error curves.

This is especially important when comparing dense and structured sketches. A structured sketch may be attractive because it is cheaper to apply, and yet it may produce larger fluctuations or require a larger sketch size before the residual and solution metrics stabilize. Conversely, a dense Gaussian sketch may look cleaner

numerically while being less attractive computationally. The experiments are designed to make this tradeoff visible.

At the present stage, this chapter should therefore be read mainly as a methodology and interpretation chapter. The benchmark structure is already concrete and implemented for least squares, and the mixed-precision side is still being expanded. Even so, the framework is already explicit enough to support the main thesis question: whether randomized compression and altered arithmetic can be studied in one coherent numerical-analysis picture.

## 5.8 Covariance-Transport Experiment Program

The later experimental layer of the thesis should test the theory from [Chapter 4  \$\rightarrow\$  p.20](#) just as directly as the least-squares benchmarks test the embedding theory from [Chapter 3  \$\rightarrow\$  p.11](#). Instead of tracking only the compressed least-squares residual, one should now measure the quality of a randomized low-rank covariance approximation and the perturbation it induces in the regularized transport map.

For a covariance operator  $C$  and its computed approximation  $\hat{C}$ , the basic numerical diagnostics should include

$$\|C - \hat{C}\|_2, \quad \|T_\lambda - \hat{T}_\lambda\|_2, \quad \|Z_f T_\lambda - Z_f \hat{T}_\lambda\|_F.$$

These quantities separate three layers of error: covariance approximation, operator perturbation of the transport map, and the induced perturbation on the corrected future data. The associated parameter sweeps should vary the target rank, the oversampling parameter, the regularization level  $\lambda$ , and the working precision used in the sketch products.

This second experimental program is also where the balancing-inexactness idea from [Chapter 2  \$\rightarrow\$  p.6](#) becomes directly testable. As the approximation rank grows, randomized truncation error should decrease. If the arithmetic regime is too aggressive, however, a point will be reached at which floating-point error no longer remains hidden beneath the approximation error. The decisive numerical question is therefore not merely whether lower precision is faster, but where the curves begin to separate and how that separation depends on regularization.

The covariance-transport experiments should therefore be organized as explicit tests rather than only as exploratory plots. A concrete first campaign is the following.

- (i) **Covariance approximation versus rank.** Fix the precision and vary the target rank  $r$  to test how quickly  $\|C - \hat{C}\|_2$  decays.
- (i) **Transport error versus regularization.** Fix the rank and vary  $\lambda$  to test how strongly the inverse-square-root conditioning affects  $\|T_\lambda - \hat{T}_\lambda\|_2$ .

- ( ) **Precision threshold sweep.** Compare double, single, and lower precision in the sketch products to identify the point at which floating-point error ceases to be dominated by randomized truncation.
- ( ) **Balancing randomized and arithmetic error.** Increase the rank until the approximation error becomes small enough that arithmetic error is no longer hidden, thereby exposing the regime predicted by the balancing principle.
- ( ) **Climate-shaped synthetic data.** Replace the purely iid Gaussian model by matrices with decaying spectra and block dependence so that the covariance-transport diagnostics are tested in a structure closer to the intended application.

Each of these experiments should be described in the final thesis by four items: the working hypothesis, the matrix or data model, the parameters being swept, and the diagnostics used to decide success or failure. That structure keeps the experimental chapter tied to the theorem statements rather than to ad hoc plotting.

The repository now contains a first concrete realization of this program in `experiments/benchmark_randmbc.R`. The current benchmark is still synthetic, but it already compares three relevant levels: the deterministic full covariance-transport baseline, selected reference methods from the vendored `MBC` package, and the current randomized `randMBC` backend under several rank, oversampling, and precision choices. The present result is not yet a speedup story. On the current synthetic data, the deterministic full baseline and the `MBC` references remain much more accurate in correlation and future-sample error than the low-rank randomized variants. This is useful evidence rather than a setback. It indicates that the dominant limitation is still the approximation regime itself, not yet the arithmetic regime.

This interpretation also sharpens the mixed-precision question. In the current benchmark, single and double precision in the randomized path are often very close to one another, while both remain far from the deterministic baseline. That is a sign that truncation and transport perturbation are still larger than the precision effect one is trying to detect. For the later experimental chapter, this means that precision sweeps should be read only after the rank and regularization regime has been pushed into a range where randomized approximation is already credible.

## 5.9 Application-Facing Diagnostics

Once the covariance-transport experiments are connected to a climate-shaped data pipeline, the numerical diagnostics should be accompanied by statistical ones. At a minimum, one should compare the corrected output against the target dependence structure through marginal quantile diagnostics, correlation diagnostics, and multivariate summaries of dependence error. Useful quantities include Pearson and Spearman correlation error, spatial and temporal autocorrelation error, and

an energy-score-style multivariate discrepancy. The point is not to replace the operator-norm viewpoint, but to relate it to the application metrics by which a post-processing method would ultimately be judged.

This distinction is important for interpretation. A method may show a visible operator perturbation while still producing acceptable application-level dependence corrections, or conversely may appear cheap and accurate in low-rank terms while failing to preserve the dependence features that matter downstream. The final experimental design should therefore keep both scales visible: the numerical scale of randomized and floating-point perturbation, and the application scale of multivariate climate diagnostics.

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